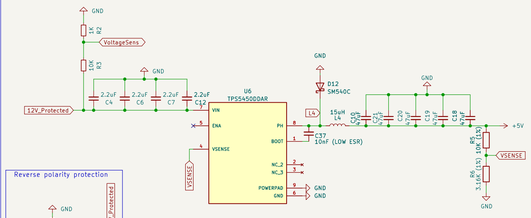
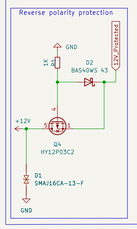
Power Regulators

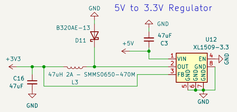
12V to 5V Regulator



Reverse polarity protection



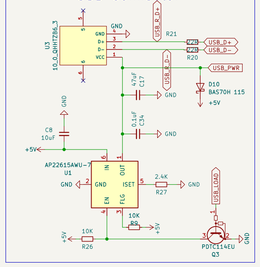
5V to 3.3V Regulator



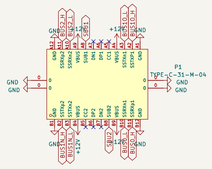
Power flags



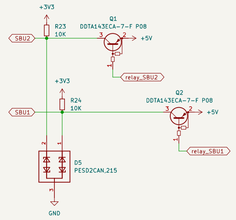
USB Connector



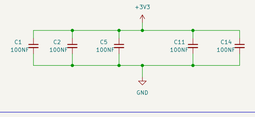
OBD-C Connector



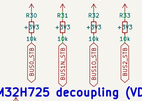
Ignition / SBU Detection



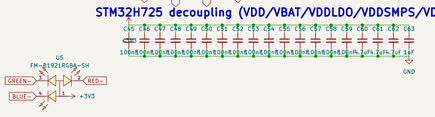
Decoupling capacitors



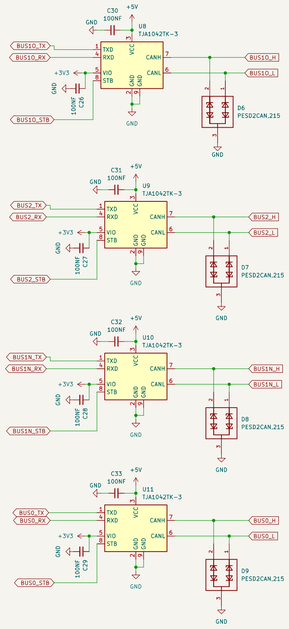
CAN STB pull-ups (default standby)



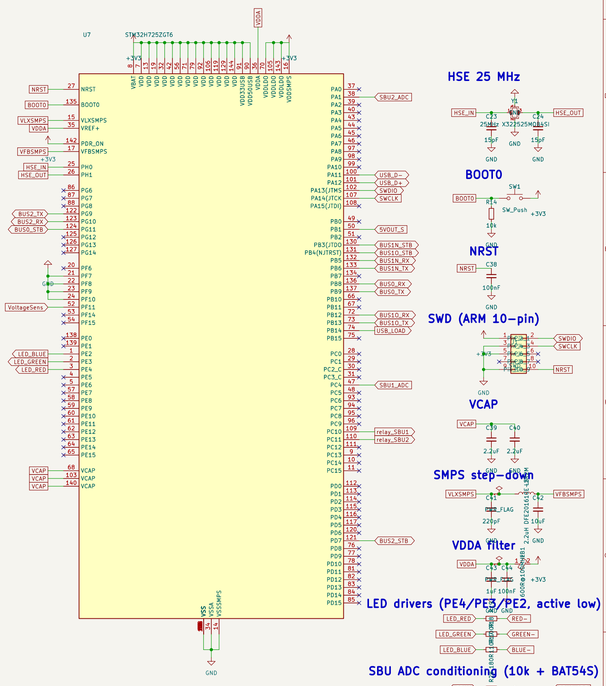
STM32H725 decoupling (VDD/VBAT/VDDLDO/VDDSMPS/VDD33USB)



CAN Interface



Microcontroller



HSE 25 MHz

BOOT0

NRST

SWD (ARM 10-pin)

VCAP

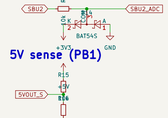
SMPS step-down

VDDA filter

LED drivers (PE4/PE3/PE2, active low)



SBU ADC conditioning (10k + BAT54S)



5V sense (PB1)